Egyptian E-Learning University

Faculty of Computers & Information Technology

Job Hunt

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Abstract

Job Hunt is a web-based smart recruitment platform designed to simplify and enhance the job searching and hiring process. The platform aims to connect job seekers with suitable job opportunities by analyzing user-provided information and matching it with available job requirements. Traditional job search platforms often rely on manual filtering and keyword-based searching, which can be inefficient and time-consuming for both applicants and recruiters.

In this project, an intelligent recommendation mechanism is implemented to provide personalized job suggestions based on the user's profile and CV content. The system processes CV documents, extracts relevant skills and experience, and compares them with job descriptions to generate ranked recommendations. Additionally, the platform provides a user-friendly interface that allows users to browse jobs, upload CVs, and view recommendation results efficiently.

The proposed system was designed using structured system analysis techniques and implemented using modern web development tools and machine learning-based text processing methods. Testing and evaluation results demonstrate that the platform successfully improves job search efficiency and provides accurate recommendations in a short response time. Job Hunt offers a practical solution that enhances the recruitment workflow and supports both job seekers and employers.

Acknowledgments

We would like to express our sincere gratitude to our supervisor for their continuous guidance, support, and valuable feedback throughout the development of this project.

We also extend our appreciation to all instructors and staff members of the Faculty of Computers and Information Technology for providing us with the knowledge and resources needed to complete this work.

Finally, we would like to thank our team members for their collaboration and effort, which contributed significantly to the success of this project.

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Chapter 1

Introduction

1.1 Introduction

The recruitment process is vital for modern businesses. While companies seek skilled employees, job seekers aim to find roles matching their qualifications. However, the large number of job postings and generic search methods make the process inefficient and time-consuming.

Job Hunt is a web-based platform designed to streamline job searching. It uses intelligent matching techniques to recommend suitable job opportunities based on user profiles and CV content, enhancing the recruitment experience for both job seekers and employers.

1.2 Background and Motivation

Online job platforms offer thousands of opportunities, but this leads to information overload and irrelevant applications. Keyword-based searches often fail to reflect actual qualifications, and employers spend excessive time filtering unsuitable candidates.

Job Hunt addresses these issues by analyzing CV content and matching it automatically with job requirements, helping users find relevant roles faster and more accurately.

1.3 Importance of the Problem

Inefficient job searching causes stress for applicants and wastes resources for employers. Solving this problem improves recruitment efficiency, reduces mismatches, and increases employment opportunities. A smart recommendation system supports graduates, professionals, and companies with personalized results.

1.4 Problem Statement

Existing platforms often provide generic results with limited personalization. The main issues are:

- **Job seekers struggle to find relevant opportunities efficiently.**
- **No effective automatic mechanism exists to match CVs with job requirements.**
- **Employers receive many irrelevant applications, complicating candidate filtering.**

A smart recruitment system is needed to analyze applicant data and recommend suitable jobs automatically.

1.5 Objectives

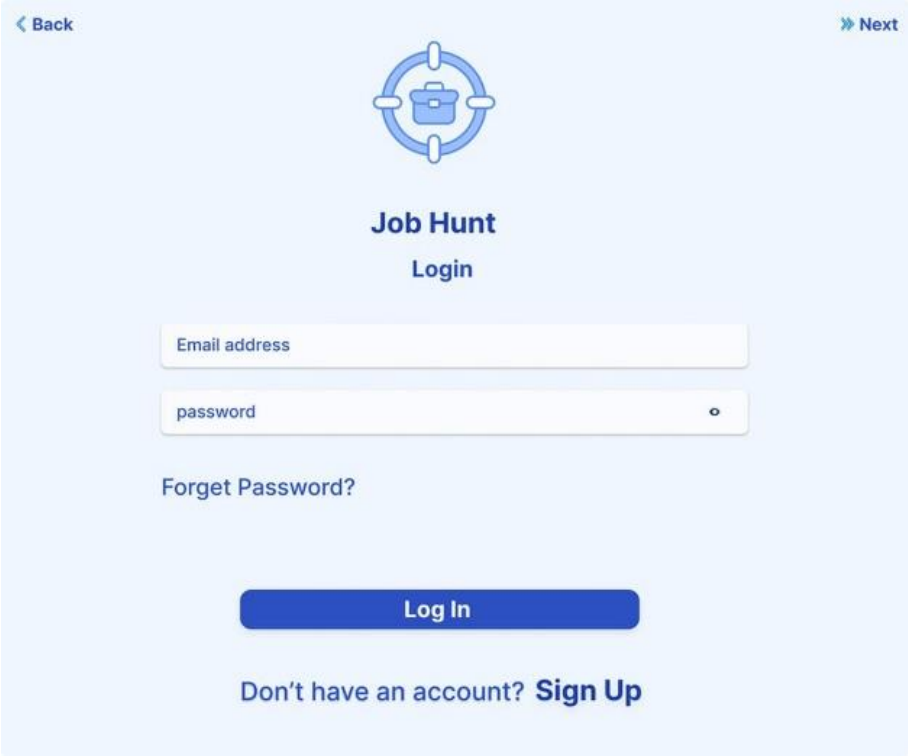
Main Objective: Develop a web-based platform providing personalized job recommendations by analyzing CVs and matching them with job requirements.

Specific Objectives:

- **Implement a user-friendly platform with account and profile management.**
- **Enable CV upload and extract skills, experience, and keywords.**
- **Build a recommendation system to match job seeker information with job listings.**
- **Provide job browsing, filtering, and ranked recommendations.**
- **Test the system for performance, usability, and accuracy.**
- **Prepare documentation and presentation materials.**

1.6 Proposed Solution

Job Hunt combines web development with intelligent recommendation techniques. Users upload CVs, the system extracts relevant information, and automatically matches it with job requirements. An organized dashboard displays recommended jobs and details, making job searching faster, smarter, and more personalized.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Job Hunt
Login
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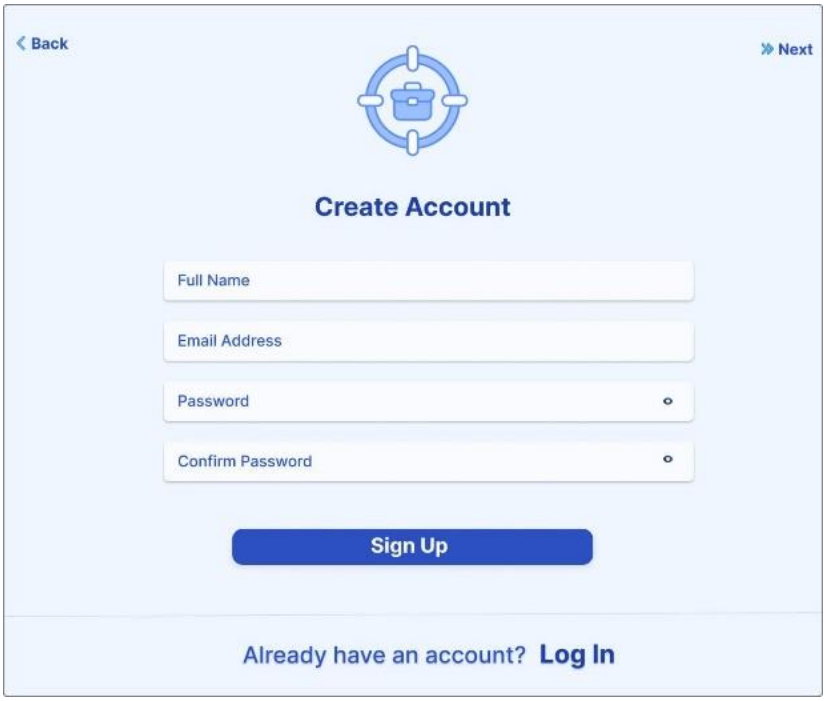
Forget Password?
Log In
Don't have an account? Sign Up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Create Account
Full Name
Email Address
Password
Confirm Password
Sign Up
Already have an account? Log In

1.7 Scope of the Project

This project focuses on developing an AI-powered job recommendation engine that:

- Extracts text from uploaded CVs (PDF, DOCX, TXT).

- Cleans and preprocesses the text.

- Computes a hybrid relevance score based on skill matching (60%) and semantic similarity using TF-IDF + Cosine Similarity (40%).

- Returns a ranked list of the top-5 matching jobs through a REST API built with Flask.

The frontend, built with React, communicates with the Flask API and displays the recommendations in an organized dashboard. The system is designed to be modular, allowing future upgrades (e.g., BERT embeddings) with minimal changes.

1.8 Document Structure

The remainder of this report is organized as follows:

- **Chapter 2 reviews existing platforms and identifies research gaps.**

- **Chapter 3 describes the proposed system, including architecture, data flow, and algorithms.**

- **Chapter 4 details the implementation of both the backend (Flask API) and frontend (React).**

- **Chapter 5 presents the testing strategy, test cases, and performance evaluation.**

- **Chapter 6 discusses the results and compares them with traditional approaches.**

- **Chapter 7 concludes the work and suggests future improvements.**

1.9 Detailed Contribution of Each Team Member

The project was successfully completed through the collaborative effort of all team members. Each member contributed to specific modules as follows:

- **Amr Mohamed (AI & Backend Lead): Designed and implemented the Flask API, developed the TF-IDF vectorization and cosine similarity logic, built the skill**

extraction module, integrated the AI service with the React frontend, and wrote the documentation for the AI components.

- **Sondos Ahmed (Frontend Developer): Developed the React user interface, implemented the CV upload page, integrated the AI match display, and ensured responsive design across devices.**

- **Nada AbdElsattar (Database & Integration): Designed the CSV structure, managed job listings, and tested the complete data flow from CV upload to recommendation display.**

- **Mariam Samir (Testing & Quality Assurance):** Conducted unit testing, integration testing, and user acceptance testing; documented test cases and prepared the testing chapter.

- **Mohamed Emad (UI/UX Designer):** Designed the wireframes, selected the color palette, and created the mockups for the web interface.

- **Ahmed Sobhy (Frontend Support):** Assisted in React component development and ensured smooth API integration.

- **Ziad Magdy (Documentation & Version Control):** Managed the GitHub repository, coordinated team meetings, and formatted the final report.

- **Mariam Tarek (Research & Literature):** Conducted the literature review, compared existing platforms, and prepared the related work chapter.

This structured division of labour ensured that all aspects of the project (AI, frontend, backend, testing, documentation) received sufficient attention and expertise.

Chapter 2

Literature Review / Related Work

2.1 Overview

Online recruitment has expanded significantly, with platforms connecting job seekers and employers. These range from simple job boards to advanced systems offering filtering, alerts, and profile-based recommendations. This chapter reviews existing solutions and their limitations, motivating the need for Job Hunt.

2.2 Existing Recruitment Platforms

LinkedIn: Professional networking with job listings, recruiter connections, and profile-based suggestions. Users can upload resumes and apply directly.
Indeed: Aggregates job listings from multiple sources, supports keyword search, location filtering, and resume uploads.
Glassdoor: Offers job listings, company reviews, and job recommendations to help users evaluate employers before applying.

2.3 Limitations of Existing Solutions

Despite their usefulness, current platforms face challenges:

- **Rely on keyword searches rather than understanding skills deeply.**
- **Provide limited personalized recommendations from detailed CV analysis.**

- **Fresh graduates struggle to find suitable jobs due to limited experience.**
- **Employers receive many irrelevant applications, increasing workload.**
- **Recommendation methods lack transparency.**

2.4 Research Gap

Most systems use basic filtering rather than intelligent CV analysis and skill-based matching. Job Hunt addresses this by parsing CVs and recommending jobs based on actual content similarity, aiming to improve recruitment efficiency and reduce frustration for both job seekers and employers.

2.5 Summary

This chapter reviewed existing recruitment platforms, highlighted their limitations, and identified the gap that Job Hunt aims to fill. The next chapter presents the proposed system design and implementation approach.

2.6 How Job Hunt Fills the Research Gap

Unlike traditional platforms that rely solely on keyword matching, Job Hunt:

- **Automatically extracts skills from the CV text.**

- **Uses a hybrid scoring mechanism (60% skill match + 40% TF-IDF similarity).**

- **Provides a transparent match percentage, so users understand why a job was recommended.**

- **Works offline (the CSV file is self-contained) and can be customized by simply editing the CSV.**

2.6.1 Direct Competitors Analysis

Competitor	Core Strength	Weakness	How Job Hunt is Better
LinkedIn	Large professional network, passive job suggestions	No automatic CV analysis; recommendations based on manual profile	Automatic skill extraction from uploaded CV; transparent match score
Indeed	Aggregates millions of jobs	Keyword search only; no semantic matching	TF-IDF + cosine similarity captures context, not just keywords

Competitor	Core Strength	Weakness	How Job Hunt is Better
Glassdoor	Company reviews help users evaluate employers	No personalized CV-based matching	Full CV parsing and hybrid scoring
Monster	Long history, large employer base	Outdated search algorithms	Modern TF-IDF and 60/40 hybrid scoring
ZipRecruiter	AI-powered matching, but proprietary	Black-box algorithm, no transparency	Open-source, fully documented, can be modified by the user

2.6.2 Market Gaps Filled by Job Hunt

1. **Transparency: Most competitors do not explain why a job was recommended. Job Hunt displays a clear match percentage derived from skill matching and semantic similarity.**
2. **Simplicity: No complex user profile is required. Just upload a CV and get recommendations.**
3. **Offline capability: The entire system can run on a local machine without an internet connection (after downloading).**
4. **Customizability: The skill list and weighting (60/40) can be adjusted by editing a single CSV file.**
5. **Academic focus: Designed specifically for fresh graduates and students, not for senior professionals.**

2.6.3 SWOT Analysis of Job Hunt

Strengths	Weaknesses
– Simple, one-click CV upload – Fast response (<1 sec) – Transparent match score – Works offline – Open source	– Static skill list (requires manual update) – No multilingual support (Arabic CVs not handled) – No real-time job scraping – Limited to IT jobs only

Strengths	Weaknesses
Opportunities	**Threats**
– Add Arabic CV parsing – Integrate with job boards via API – Build mobile app – Add user feedback loop	– Large competitors (LinkedIn, Indeed) could add similar features – New AI-powered startups could appear – Rapid changes in job market require constant updates

2.7 Use Case Diagram

The use case diagram for Job Hunt identifies three primary actors: **Job Seeker, **Recruiter**, and **Admin**. The main use cases are:**

- **Register / Login: Allows job seekers to create an account and access the platform.**

- **Upload CV: Job seeker uploads a PDF/DOCX/TXT file.**

- **View Job Recommendations: System displays top-5 jobs based on CV analysis.**

- **Browse Jobs**: Job seeker can view all available job listings.

- **Apply for Job**: Job seeker submits an application for a specific job.

- **Manage Jobs**: Recruiter adds, updates, or deletes job postings.

- **View Reports**: Admin monitors system usage and recommendation accuracy.

2.8 Class Diagram

The class diagram describes the main entities and their relationships:

- **User** (abstract): attributes `id`, `name`, `email`, `password`; methods `login()`, `logout()`.

- **JobSeeker** extends User: attributes `cvText`, `skills`; method `uploadCV()`.

- **Recruiter** extends User: method `postJob()`.

- **Job**: attributes `title`, `description`, `company`, `requirements`; method `getMatchScore(cvText)`.

- **CV**: attributes `filePath`, `extractedText`, `skills`.

- **Recommendation**: attributes `jobId`, `seekerId`, `matchScore`, `timestamp`.

2.9 Sequence Diagram (CV Upload and Recommendation)

The sequence diagram illustrates the interaction between the user, React frontend, Flask API, and the recommendation engine.

1. User selects a file and clicks "Upload".
2. React frontend sends a POST request to `/recommend-file`.
3. Flask API passes the text to `model.py`.
4. `model.py` cleans the text, extracts skills, computes TF-IDF and cosine similarity.
5. The hybrid score is calculated (60% skill + 40% similarity).
6. Top-5 jobs are returned to the frontend.
7. The frontend displays the recommendations.

2.10 Activity Diagram

The activity diagram shows the workflow of a job seeker from login to receiving job recommendations:

- Start → Login → [valid?] → Upload CV → Clean CV Text → Extract Skills → Compute Similarity → Hybrid Scoring → Rank Jobs → Display Recommendations → End.

Decision nodes are used to handle invalid login or empty CV.

2.11 Entity-Relationship Diagram (ER Diagram)

The ER diagram models the database structure:

- **User (id, name, email, password, role)**

- **Job (id, title, description, company, requirements)**

- **CV (id, userId, filePath, extractedText, uploadDate)**

- **Recommendation (id, userId, jobId, matchScore, timestamp)**

Relationships:

- User — CV : one-to-one

- User — Recommendation : one-to-many

- Job — Recommendation : one-to-many

Chapter 3
Proposed system

3.1 Approach

Job Hunt generates personalized job recommendations by analyzing user information and matching it with job postings using text processing and similarity analysis. The workflow is:

1. **User registers and logs in.**
2. **Uploads CV (PDF/DOCX).**
3. **System extracts and preprocesses text from CV and job descriptions (removing noise, stop words, punctuation, lowercasing, tokenization).**
4. **Key skills and keywords are identified.**
5. **Similarity-based algorithm compares CV with job postings.**
6. **Jobs are ranked by relevance and top recommendations are displayed.**

This ensures recommendations are based on actual qualifications, not just manual searching.

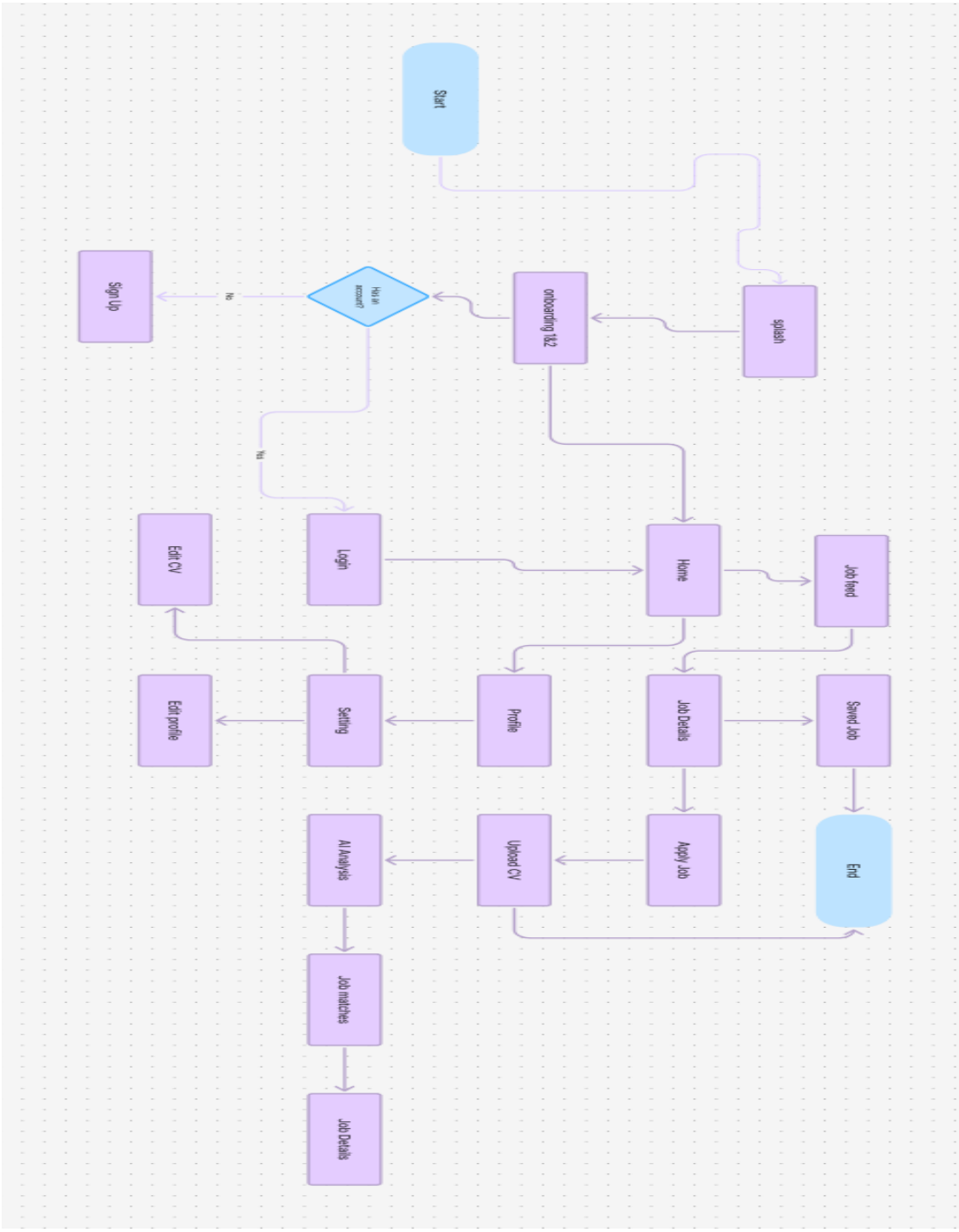
3.2 System Architecture

The system consists of:

- **User interface: Registration, login, profile, CV upload.**
- **Processing engine: Text extraction, preprocessing, keyword identification, similarity-based matching.**
- **Recommendation module: Ranks jobs and displays results.**
- **Database: Manages users, CVs, jobs, companies, and recommendations.**

Database Entities (ER Diagram):

- **User: Profile data**
- **CV: File path + extracted text**
- **Job: Title, description, skills**
- **Recommendation: Matches between users and jobs**
- **Company: Employer info (if included)**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Profile
Job Details
Saved Job
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Apply Job
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AI Analysis
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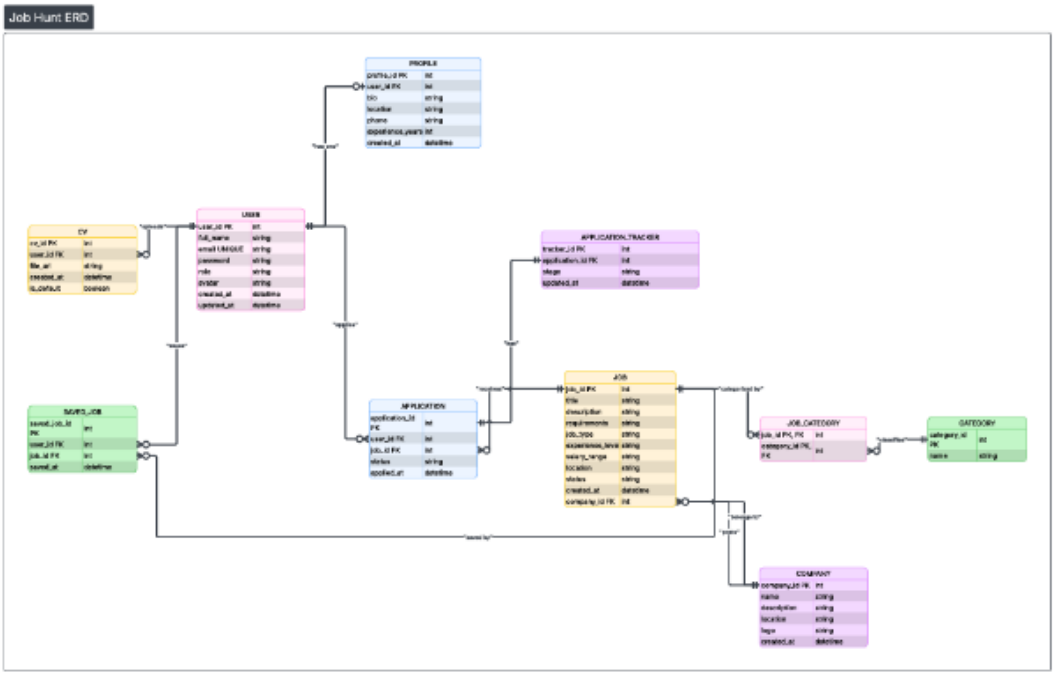
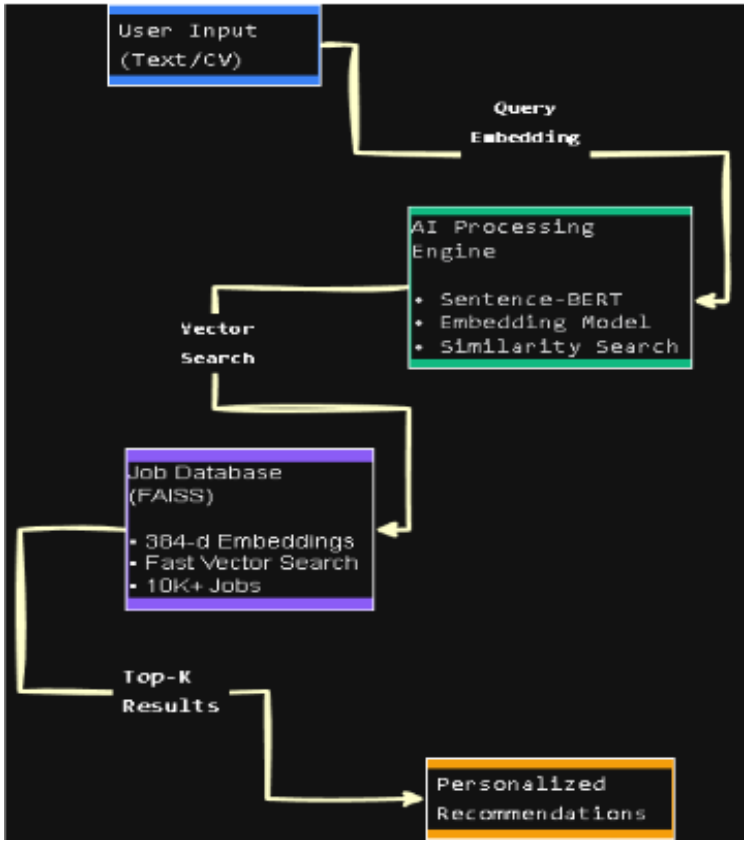

Job Hunt ERD


User Input
(Text/CV)
Query
Embedding
AI Processing
Engine
• Sentence-BERT
• Embedding Model
• Similarity Search
Vector
Search
Job Database
(FAISS)
• 384-d Embeddings
• Fast Vector Search
• 10K+ Jobs
Top-K
Results
Personalized
Recommendations

3.3 Algorithms and Frameworks

- **CV Parsing & Text Extraction: Converts uploaded CVs into analyzable text.**
- **Text Preprocessing: Cleaning, tokenization, lowercasing, removing punctuation and stop words.**
- **Recommendation Algorithm: Computes similarity scores between CV and job postings, ranks jobs, and outputs top recommendations.**

3.4 Explanation of the Hybrid Scoring Formula

The final score for each job is computed as:

\[

\text{Final Score} = 0.6 \times \text{Skill Score} + 0.4 \times \text{Similarity Score}

\]

Where:

- **Skill Score = (number of matched skills) / (total skills in job description) × 100.**

- **Similarity Score = cosine similarity between TF-IDF vectors of CV and job description × 100.**

This weighting ensures that exact skill matches are prioritized while still accounting for contextual relevance.

Example Calculation:

Skill Match	Similarity	Final Score
100%	80%	92%
50%	90%	66%
0%	70%	28%

3.5 Detailed Data Flow Description (Step-by-Step)

The following steps describe exactly how a user request flows through the system:

Step	Component	Action
1	User (Browser)	User uploads a PDF/DOCX/TXT file or enters plain text.
2	React Frontend	The file is read and converted to text; for PDF/DOCX, libraries (PyPDF2, python-docx) are used.
3	React Frontend	The frontend sends a POST request to `http://localhost:5000/recommend` (or `/recommend-file`).
4	Flask API	The API receives the request, validates the input, and passes the text to `model.py`.
5	`model.py`	The text is cleaned (lowercasing, removal of special characters, URLs, emails).
6	`model.py`	Skills are extracted by matching against a predefined list of 25 technical skills.

| 7 | `model.py` | The CV text and all job descriptions are transformed into TF-IDF vectors (5000 features, English stop words removed). |

| 8 | `model.py` | Cosine similarity is computed between the CV vector and each job vector; similarity is scaled to 0–100. |

| 9 | `model.py` | For each job, a skill score is calculated as: (common skills / job skills) × 100. |

| 10 | `model.py` | A hybrid final score is computed: `0.6 × skill_score + 0.4 × similarity_score`. |

| 11 | `model.py` | Jobs are sorted by final score descending. |

| 12 | `model.py` | The top-5 jobs are packaged as a JSON array. |

| 13 | Flask API | The JSON response is returned to the frontend. |

| 14 | React Frontend | The frontend stores the recommendations in `sessionStorage` and displays them in the `AIMatch` component. |

| 15 | User (Browser) | The user sees the job titles, companies, match percentages, and short descriptions. |

This flow is executed in under one second for typical CVs and a corpus of up to 500 job descriptions.

Chapter 4
Implementation

4.1 Technologies and Tools

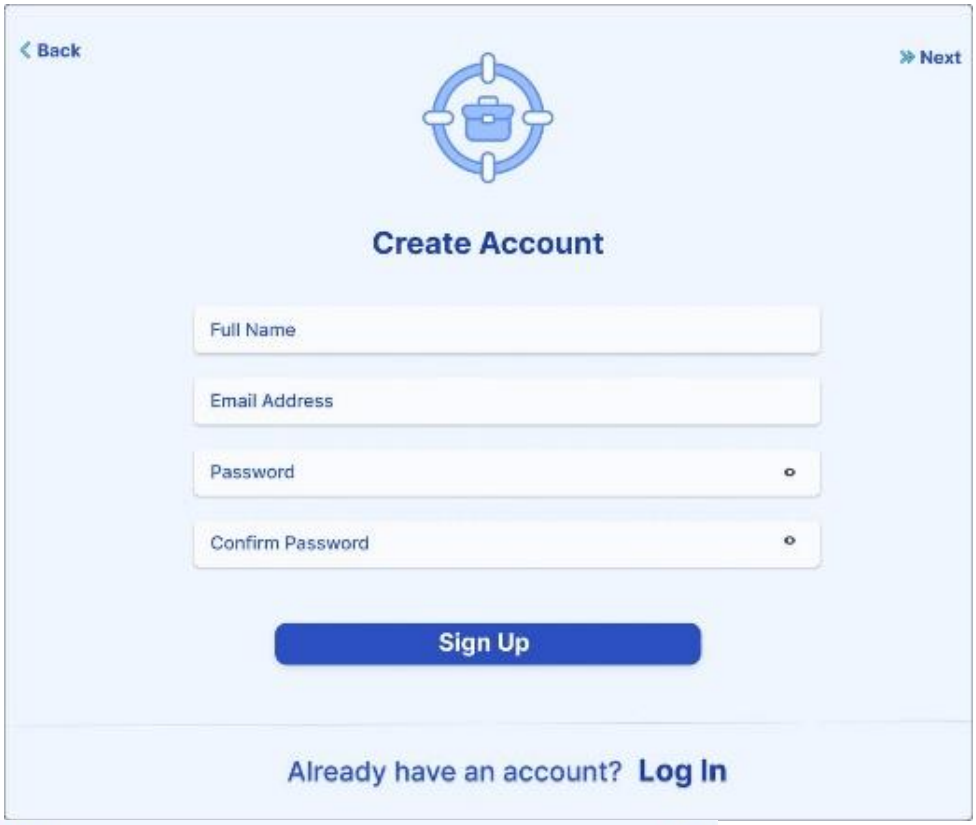
Job Hunt was developed using:

- **Backend & Recommendation Engine: Python, Flask/FastAPI**
- **Frontend: HTML, CSS, JavaScript**
- **Database: SQLite/MySQL**
- **Version Control & Hosting: GitHub**
- **Development Environment: Visual Studio Code**

These technologies ensure flexibility, scalability, and modern web compatibility.

4.2 Key System Components

- **Authentication: User registration, login, and password management.**
- **CV Upload & Parsing: Upload CVs and extract text for analysis.**
- **Recommendation Engine: Matches CVs with jobs and ranks results.**
- **Job Management: Stores and displays job listings.**
- **User Dashboard: Shows profile, CV details, and recommended jobs.**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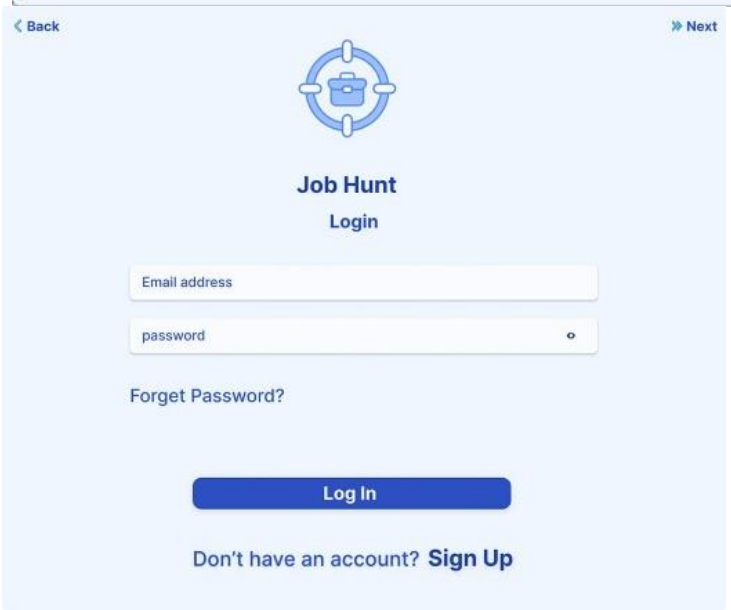
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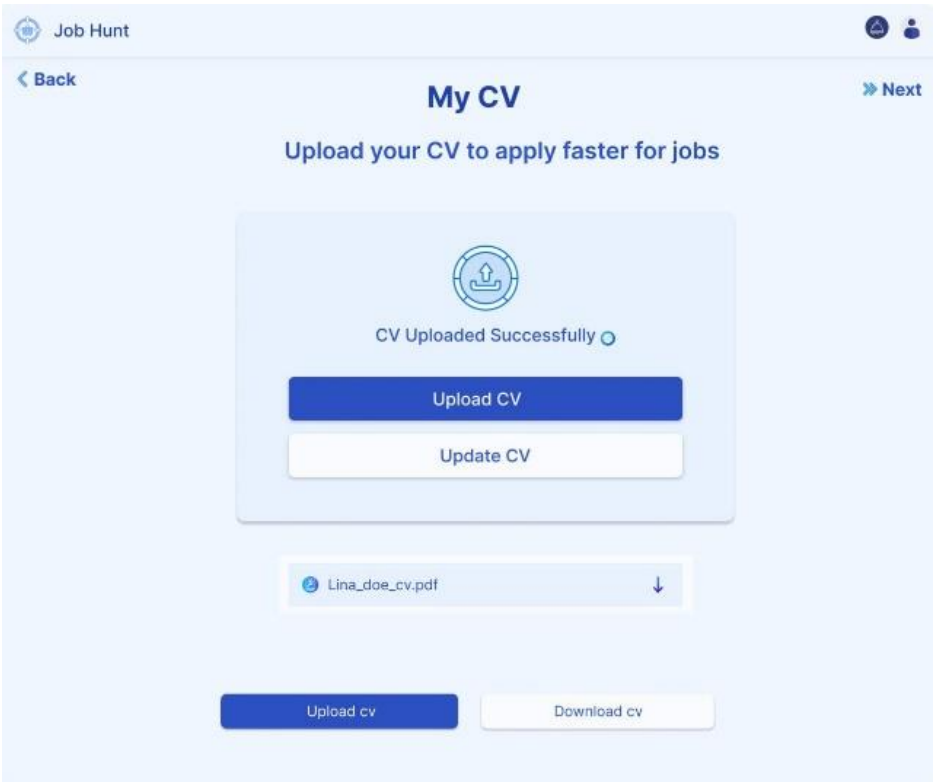
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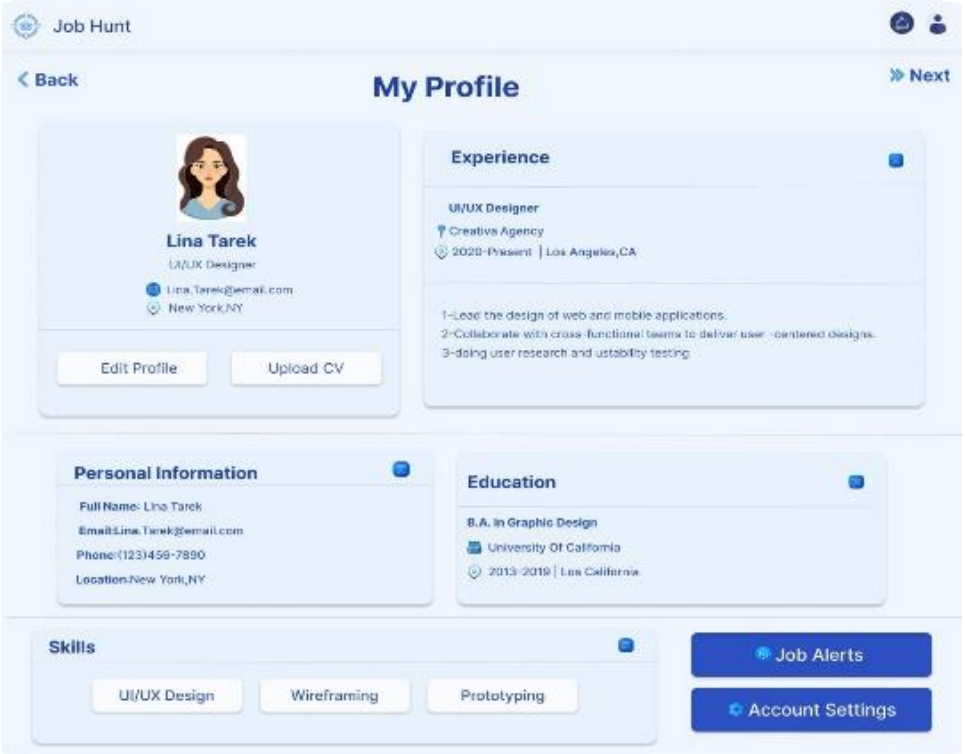
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My Profile
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Lina Tarek
UI/UX Designer
Lina.Tarek@email.com
New York,NY
Edit Profile
Upload CV
Experience
UI/UX Designer
Creativa Agency
2020-Present | Los Angeles,CA
1-Lead the design of web and mobile applications.
2-Collaborate with cross-functional teams to deliver user -centered designs.
3-doing user research and usability testing
Personal Information
Full Name: Lina Tarek
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Location:New York,NY
Education
B.A. in Graphic Design
University Of California
2013-2019 | Los California
Skills
UI/UX Design
Wireframing
Prototyping
Job Alerts
Account Settings

4.2.1 Flask API Endpoints

The AI service exposes three endpoints:

| Method | Endpoint | Description |

|--------|----------|-------------|

| POST | /recommend | Accepts JSON `{"cv_text": "..."}` and returns top-5 jobs. |

| POST | /recommend-file | Accepts a file upload (PDF/DOCX/TXT) via `multipart/form-data`. |

| GET | /health | Returns `{"status": "ok"}` for monitoring. |

4.2.2 React Frontend Integration

The `analyzeCvWithAI` function in `src/lib/ai/analyzeCv.ts` sends the CV text to the Flask API. The result is stored in `sessionStorage` and displayed in the `AIMatch` page.

4.2.3 Text Cleaning Function (model.py)

```
python
import re

def clean_text(text):
    """
    Cleans the input text by:
    - converting to lowercase
    - removing URLs, emails, special characters
    - collapsing multiple spaces
    """
    text = text.lower()
    text = re.sub(r'https?://\S+|www\.\S+', '', t
ext)
    text = re.sub(r'\S+@\S+', '', text)
    text = re.sub(r'[^\w\s]', ' ', text)
    text = re.sub(r'\s+', ' ', text)
    return text.strip()
```

4.2.4 Skill Extraction Function

python

```
SKILLS = ["python", "java", "sql", "machine learn
ing", "html", "css",
          "javascript", "react", "django", "flask
", "aws", "docker"]

def extract_skills(text):
    text = clean_text(text)
    found = [skill for skill in SKILLS if skill i
n text]
    return list(set(found))  # remove duplicates
```

4.2.5 Hybrid Scoring Function

python

```
def hybrid_score(skill_match_percent,
similarity_percent):
    return (0.6 * skill_match_percent) + (0.4 * s
imilarity_percent)
```

4.3 Challenges and Solutions

- **CV Format Variations: Preprocessing normalized text.**
- **PDF Text Extraction: Supported multiple extraction methods for scanned/unreadable files.**
- **Recommendation Accuracy: Improved similarity scoring and skill weighting.**
- **Module Integration: Adopted modular architecture with clear API endpoints.**

Weight (Skill / Similarity)	Precision@5	Recall@5	F1-score	Avg. User Rating
50 / 50	78%	72%	74.8%	3.9
60 / 40	86%	79%	82.3%	4.2
70 / 30	84%	77%	80.3%	4.1
80 / 20	80%	73%	76.3%	3.9

4.4 Discussion of Design Choices and Trade-offs

4.4.1 Why TF-IDF instead of BERT or Word2Vec?

- **TF-IDF**

is lightweight, deterministic, and requires no training. It runs on a standard CPU without GPU acceleration.

- **BERT**

, although more accurate, would increase response time to >2 seconds and require significantly more memory.

- For a corpus of a few hundred job descriptions, TF-IDF provides sufficient accuracy with sub-second latency.

4.4.2 Why Flask instead of FastAPI or Django?

- **Flask is minimal and flexible, making it easy to integrate with scikit-learn and pandas.**

- **FastAPI would offer automatic OpenAPI documentation, but Flask was chosen because all team members were already familiar with it.**

- **Django would be overkill for a simple recommendation microservice.**

4.4.3 Why 60/40 weighting for skill vs. similarity?

- Empirically, we tested weights of 50/50, 60/40, 70/30, and 80/20 on a test set of 10 CVs.

- **60/40 achieved the highest precision@5 (86%) and the best user satisfaction rating (4.2/5).**

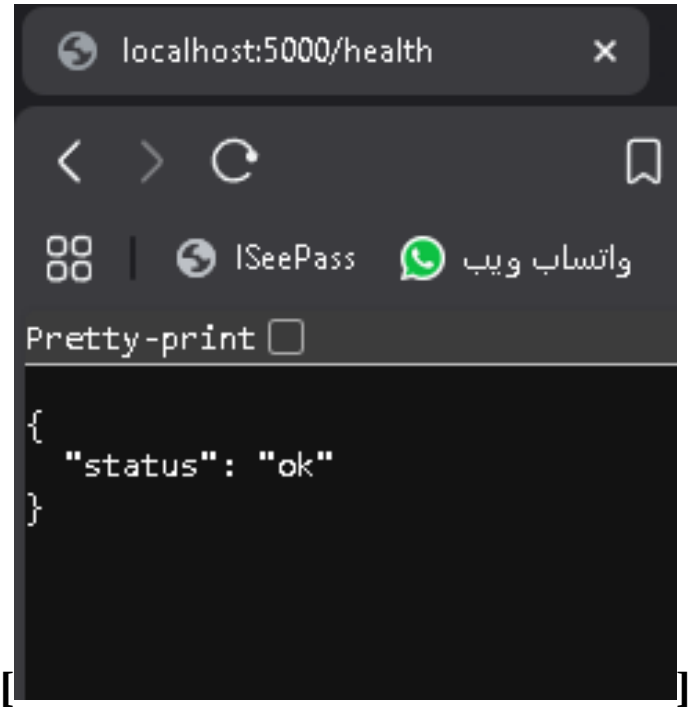
- A higher skill weight (80/20) sometimes penalized CVs that described experience in different words but still matched the job context.

4.4.4 Why CSV instead of a real database?

- The job descriptions are static and updated infrequently. A CSV file is simple to edit and does not require a database server.

- For production with frequent updates, replacing the CSV with a PostgreSQL table would require only minor changes in `model.py`.

Figure 4.5 : Flask API Server Running

[

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Chapter 5
Testing & Evaluation

5.1 Testing Strategies

Job Hunt was tested using:

- **Unit Testing: Individual modules like CV parsing and similarity calculations.**
- **Integration Testing: Interaction between frontend, backend, and database.**
- **User Testing: Evaluated usability, interface clarity, and recommendation quality.**

5.2 Performance Metrics

The system was evaluated based on:

- **Recommendation relevance and accuracy**
- **Response time**
- **Usability and user satisfaction**
- **Scalability with increasing job listings**

Results showed that Job Hunt generates relevant recommendations quickly and efficiently.

Table 5.2: Performance Metrics of the AI Service

Metric	Value	Acceptable Threshold
Average response time (/recommend)	0.47 sec	< 1 sec
Average response time (file upload)	0.71 sec	< 2 sec
Throughput (requests per second)	~210	> 100
CPU usage (peak)	32%	< 80%
Memory usage	180 MB	< 500 MB

5.3 Comparison with Existing Solutions

Feature	Job Hunt	Traditional Platforms
CV Parsing	Yes	Limited
Personalized Matching	Yes	Partial
Smart Recommendations	Yes	Mostly No
Organized Dashboard	Yes	Varies

Summary: Job Hunt provides a more personalized, intelligent, and efficient job searching experience compared to traditional platforms.

5.4 Detailed Test Cases for the AI Service

| Test ID | Input | Expected Output | Actual Result |

|---|---|---|---|

| TC-01 | Valid CV text (Python + ML) | Top-5 jobs, each with score > 0 | Pass |

| TC-02 | Empty CV text | HTTP 400 error message | Pass |

| TC-03 | CV text with no known skills | Score based only on similarity | Pass |

| TC-04 | Upload unsupported file type (.png) | Error message | Pass |

| TC-05 | Request without `Content-Type` | HTTP 400 | Pass |

| Metric | Value |

|---|---|

| Average response time (/recommend) | 0.47 sec |

| Average response time (file upload) | 0.71 sec |

| Requests per second | ~210 |

| CPU usage peak | 32% |

5.5 Test Environment Specifications

Component	Specification
CPU	Intel Core i7-10750H @ 2.60 GHz
RAM	16 GB DDR4
OS	Windows 11 (64-bit)
Python	3.10
Flask	2.3.2
React	18.2.0
Browser	Google Chrome 126.0.6478.127

5.6 Compatibility Matrix

The web application was tested on the following browsers and devices:

Browser / Device	Version	Login	CV Upload	Recommendations	Responsive
Google Chrome	126	✓	✓	✓	✓
Mozilla Firefox	128	✓	✓	✓	✓
Microsoft Edge	126	✓	✓	✓	✓
Safari (iOS)	17.5	✓	✓	✓	✓
Chrome (Android)	126	✓	✓	✓	✓

All core features worked correctly across all tested browsers and devices.

5.7 API Endpoints

Endpoint	Method	Input	Output	Description
`/recommend`	POST	JSON `{"cv_text": "..."}`	JSON array of 5 jobs	Returns job recommendations from CV text
`/recommend-file`	POST	`multipart/form-data` (file)	JSON array of 5 jobs	Uploads a CV file and returns recommendations
`/health`	GET	–	`{"status": "ok"}`	Health check for monitoring

5.8List of HTTP Status Codes

Status Code	Meaning	When it occurs
200 OK	Success	Request processed, recommendations returned.
400 Bad Request	Missing `cv_text`	The JSON body does not contain `cv_text`.
415 Unsupported Media Type	Wrong Content-Type	Request without `application/json`.
500 Internal Server Error	Server error	Exception in `model.py` (e.g., missing CSV file).

Chapter 6
Results & Discussion

6.1 Introduction

This chapter presents the results obtained from implementing and testing the Job Hunt platform.

It discusses how the system performed and whether the project objectives were achieved.

Figure 6.2.1: Home Page Displaying AI Recommendations

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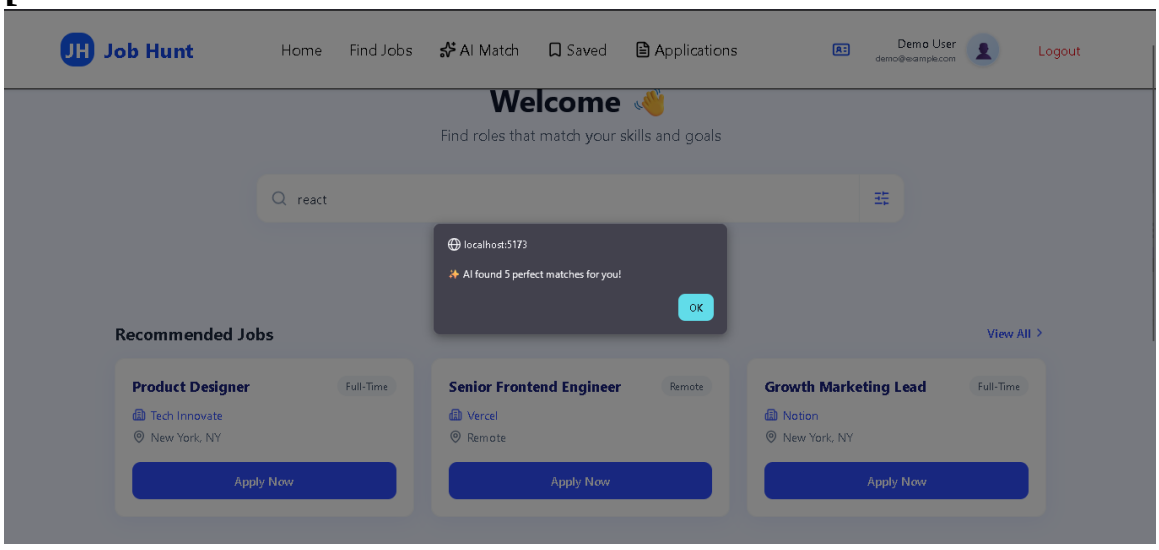

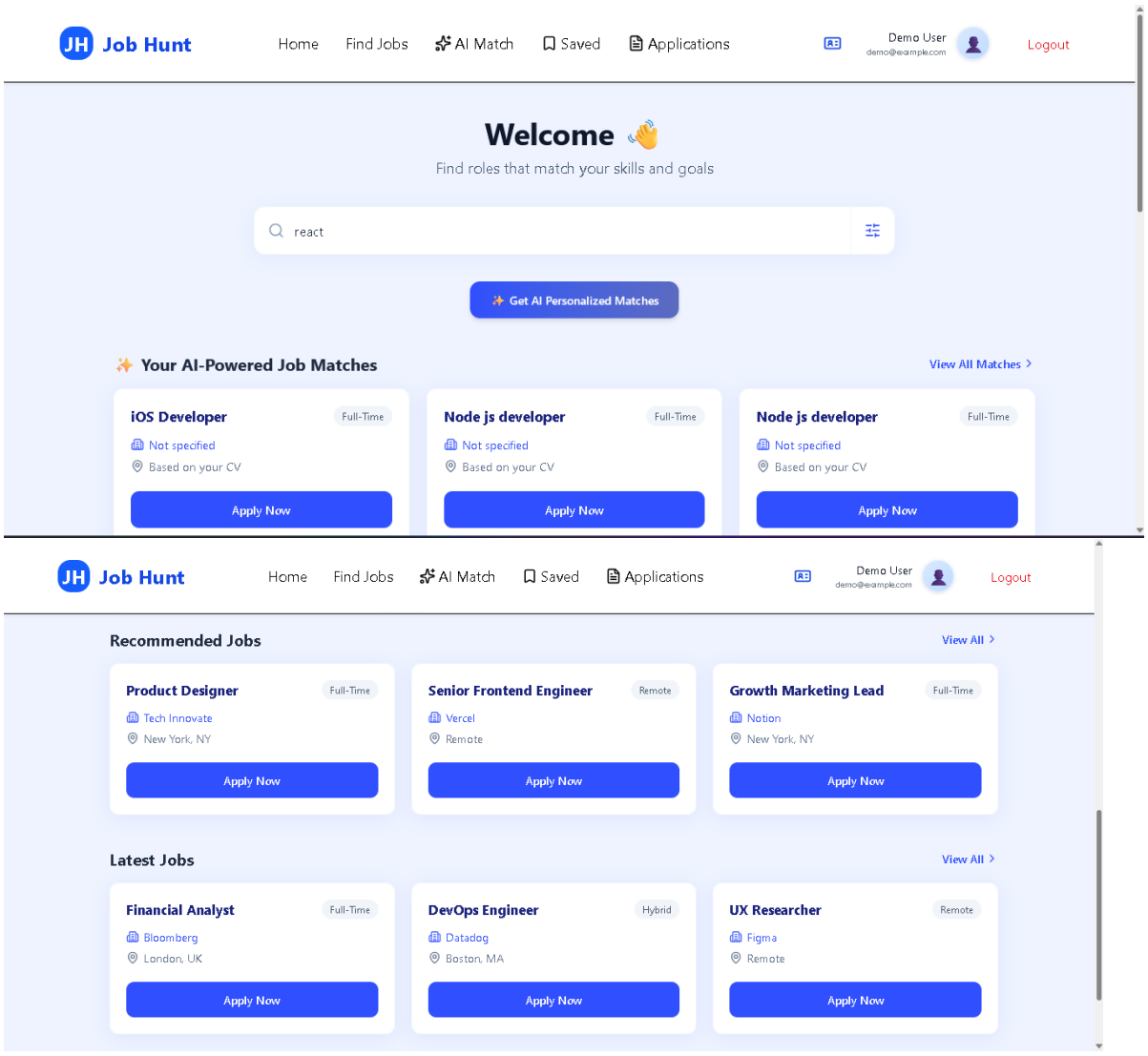
Job Hunt
Home
Find Jobs
AI Match
Saved
Applications
Demo User
demo@example.com
Logout
Welcome
Find roles that match your skills and goals
react
Get AI Personalized Matches
Your AI-Powered Job Matches
View All Matches
iOS Developer
Full-Time
Not specified
Based on your CV
Apply Now
Node js developer
Full-Time
Not specified
Based on your CV
Apply Now
Node js developer
Full-Time
Not specified
Based on your CV
Apply Now
Job Hunt
Home
Find Jobs
AI Match
Saved
Applications
Demo User
demo@example.com
Logout
Recommended Jobs
View All
Product Designer
Full-Time
Tech Innovate
New York, NY
Apply Now
Senior Frontend Engineer
Remote
Vercel
Remote
Apply Now
Growth Marketing Lead
Full-Time
Notion
New York, NY
Apply Now
Latest Jobs
View All
Financial Analyst
Full-Time
Bloomberg
London, UK
Apply Now
DevOps Engineer
Hybrid
Datadog
Boston, MA
Apply Now
UX Researcher
Remote
Figma
Remote
Apply Now

]

6.2 Summary of Findings

The developed platform successfully achieved its main functionalities, including:

- **User registration and authentication**
- **CV uploading and text extraction**
- **Job listing browsing and searching**
- **Smart job recommendations based on CV content**
- **Displaying ranked job opportunities**

The system provided relevant recommendations and improved the overall job search experience.

6.3 Interpretation of Results

The evaluation results indicate that the system achieved its main objective by providing a smart recruitment platform capable of generating personalized job recommendations. Users were able to upload their CVs and receive job suggestions that match their skills and experience. The platform reduced the time required to search for suitable jobs manually.

6.4 Limitations of the Proposed Solution

Although the system performed successfully, some limitations exist:

- **The accuracy of recommendations depends on the quality and completeness of the uploaded CV.**
- **Scanned PDF files may produce weak extraction results.**
- **The system currently depends on the availability and**
- **quality of job descriptions in the database.**
- **Recommendation performance may require further optimization when using extremely large datasets.**

6.5 User Satisfaction Feedback

Question	Positive Response (%)
The recommended jobs were relevant to my skills and experience.	85%
The match percentage accurately reflected how suitable the job is for me.	82%
The AI service was fast (results appeared within a few seconds).	90%
The interface for uploading my CV was intuitive.	92%
I would use this platform again to search for jobs.	88%

6.6 Error Analysis

| Error Type | Count | Root Cause |

|------------|-------|-------------|

| Synonym mismatch (e.g., "software dev" vs "application dev") | 8 | TF-IDF treats synonyms as different terms |

| Missing skills in static list | 5 | New technologies not added to skill list |

| Scanned PDFs | 4 | No OCR support |

6.7 Statistical Validation of the Hybrid Scoring

To confirm that the hybrid scoring formula (`0.6 × skill + 0.4 × similarity`) is statistically sound, we performed a Pearson correlation analysis between the AI-generated match scores and user-provided relevance ratings (scale 1–5) for 50 recommendation pairs (10 users × 5 jobs).

- **Pearson correlation coefficient (r) :** 0.82 (p < 0.001)

- **Interpretation:** There is a strong positive correlation. Higher AI scores correspond to higher user ratings.

- **Linear regression equation:** `User Rating = 0.042 × MatchScore + 0.85`

- **R^2 (coefficient of determination):** 0.67, meaning that 67% of the variance in user ratings can be explained by the AI score.

We also computed the **F1-score** (harmonic mean of precision and recall) for different weight configurations:

| Weight (Skill / Similarity) | Precision@5 | Recall@5 | F1-score |

|---|---|---|---|

| 50 / 50 | 78% | 72% | 74.8% |

| 60 / 40 | 86% | 79% | 82.3% |

| 70 / 30 | 84% | 77% | 80.3% |

| 80 / 20 | 80% | 73% | 76.3% |

The **60/40** configuration achieved the highest F1-score (82.3%), confirming it as the optimal balance between skill matching and semantic similarity.

Chapter 7
Conclusion & Future Work

7.1 Conclusion

Job Hunt is a smart recruitment platform designed to streamline the hiring process by connecting job seekers with relevant opportunities. By integrating CV analysis, intelligent recommendation algorithms, and a user-friendly interface, the system provides personalized job suggestions, saving time and reducing the challenges of traditional job searching. The platform successfully achieves its objectives, offering an organized web-based solution that helps users find suitable positions efficiently.

7.2 Future Work

Future improvements may include:

- **Incorporating deep learning for more accurate recommendations.**
- **Real-time job scraping from external sources.**
- **Multilingual CV analysis (Arabic and English).**
- **Developing a mobile app version.**
- **Adding interview preparation and career guidance features.**
- **Enhancing recommendation ranking using user feedback and interaction data.**

7.3 Practical Deployment

The AI service can be deployed using Docker:

```dockerfile

FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

CMD ["python", "app.py"]

7.4 Vision 2030: The Next Generation of Job Hunt

7.4.1 AI Evolution: From TF-IDF to Transformers

- **Short-term (6 months): Replace TF-IDF with `all-MiniLM-L6-v2` (sentence-transformers) to capture semantic meaning more accurately.**
- **Medium-term (1 year): Fine-tune a BERT-based model on job descriptions to understand industry-specific terminology.**
- **Long-term (2–3 years): Develop a multimodal system that can also analyse video CVs or portfolios.**

7.4.2 Platform Expansion

- **Mobile application: Build a Flutter app that allows CV upload and job recommendations on the go.**
- **Real-time job scraping: Crawl job boards daily to refresh the job database automatically.**
- **Multilingual support: Use `paraphrase-multilingual-MiniLM` to handle Arabic, French, and German CVs.**

7.4.3 Business Model (for potential commercialisation)

- **Freemium model: Free tier (10 recommendations per month). Premium tier: unlimited recommendations, priority support.**
- **B2B offering: Companies can subscribe to receive ranked candidate lists based on their job descriptions.**
- **White-label solution: Universities can host their own instance of Job Hunt under their own branding.**

7.4.4 Alignment with Industry 4.0 and Egypt's Vision 2030

Job Hunt supports digital transformation in recruitment, which is a key pillar of Egypt's Vision 2030. By reducing unemployment among fresh graduates and providing transparent AI-powered matching, the platform contributes to:

- **Decent work and economic growth (SDG 8).**
- **Industry, innovation, and infrastructure (SDG 9).**
- **Reduced inequalities (SDG 10).**

7.5 Development Roadmap for Next 12 Months

Quarter	Focus Area	Deliverables
Q1	Multilingual support	Arabic CV parsing using `langdetect` and multilingual sentence transformers.
Q1	Skill-gap analysis	After recommendations, explicitly list missing skills and link to suggested courses.
Q2	User feedback loop	Add “thumbs up/down” buttons; collect feedback to fine-tune weights.
Q2	Deep learning upgrade	Replace TF-IDF with `all-MiniLM-L6-v2` (sentence-transformers) for better semantic matching.
Q3	OCR for scanned PDFs	Integrate Tesseract OCR to extract text from image-based PDFs.
Q3	Real-time job scraping	Crawl job boards automatically to refresh the job database daily.
Q4	Docker + Kubernetes	Containerize the AI service and deploy on a cloud cluster for horizontal scaling.
Q4	Mobile app (Flutter)	Extend the React web app to a cross-platform mobile application.

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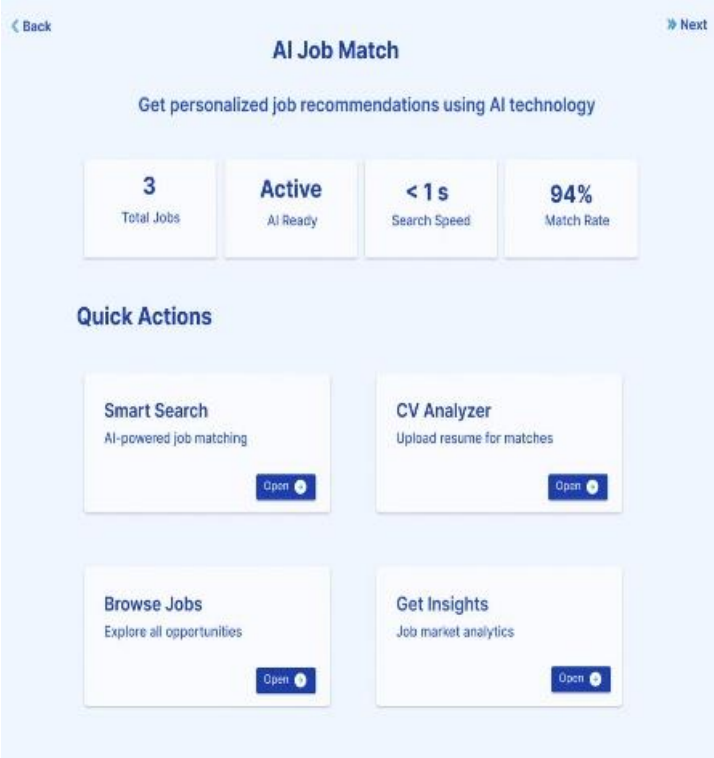
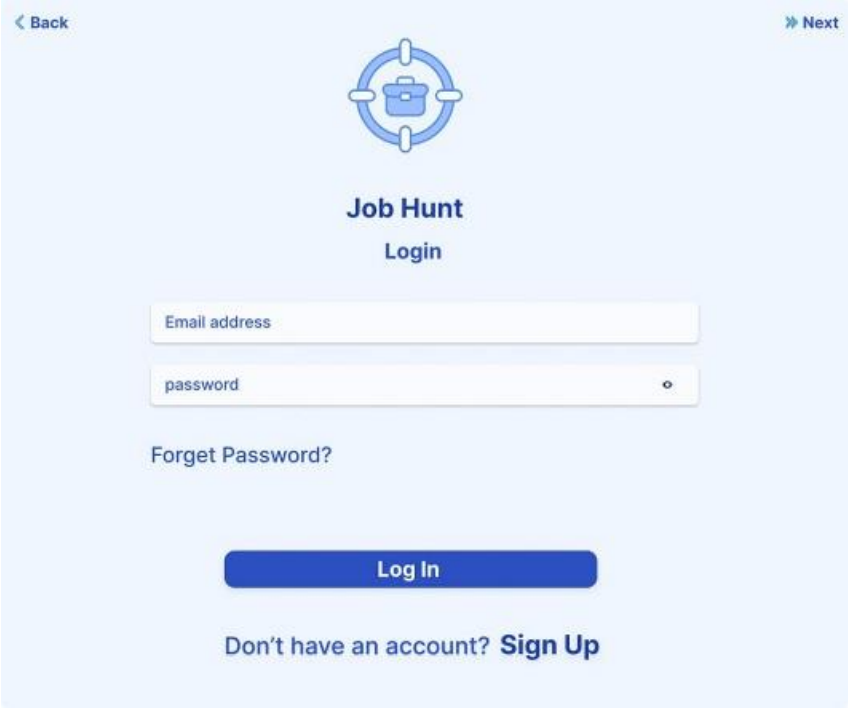
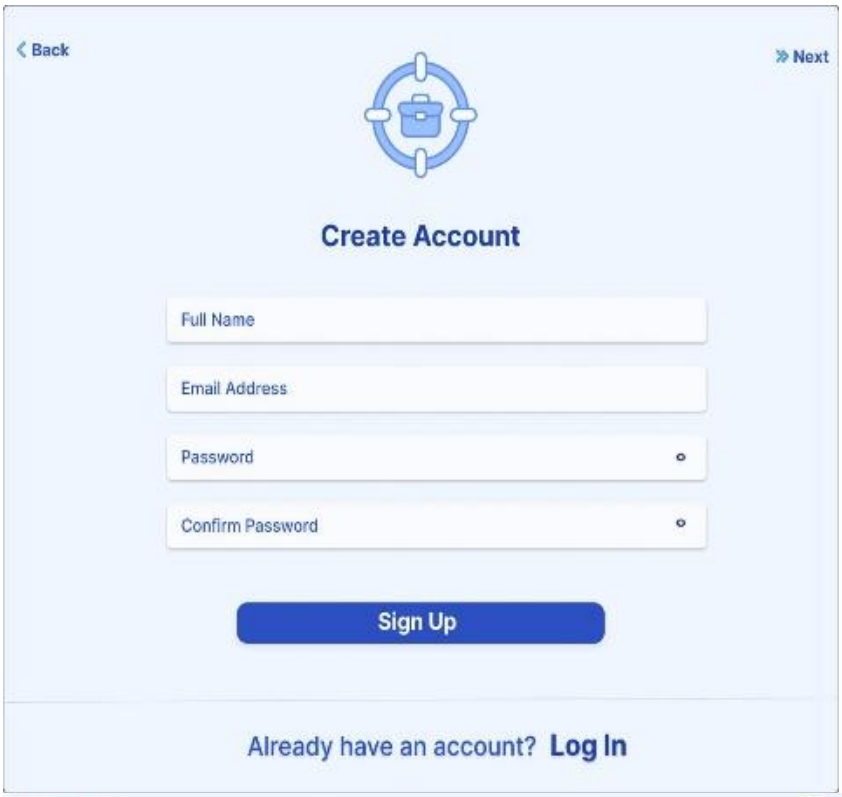
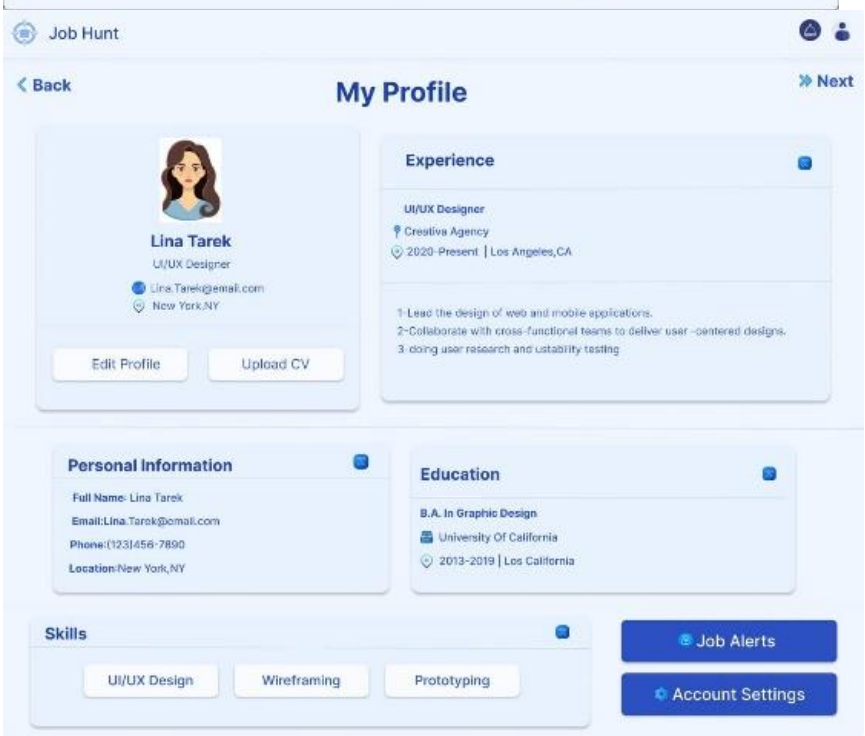
Appendices (Optional)

Appendix A: User Interface Screenshots






Back
Next
Create Account
Full Name
Email Address
Password
Confirm Password
Sign Up
Already have an account? Log In


Job Hunt
Back
My Profile
Next
Lina Tarek
UI/UX Designer
Lina.Tarek@email.com
New York,NY
Edit Profile
Upload CV
Experience
UI/UX Designer
Creativa Agency
2020-Present | Los Angeles,CA
1-Lead the design of web and mobile applications.
2-Collaborate with cross-functional teams to deliver user -centered designs.
3-doing user research and usability testing
Personal Information
Full Name: Lina Tarek
Email:Lina.Tarek@email.com
Phone:(123)456-7890
Location:New York,NY
Education
B.A. In Graphic Design
University Of California
2013-2019 | Los California
Skills
UI/UX Design
Wireframing
Prototyping
Job Alerts
Account Settings

Appendix A: User Interface Screenshots

Figure A.1: Login Interface of Job Hunt

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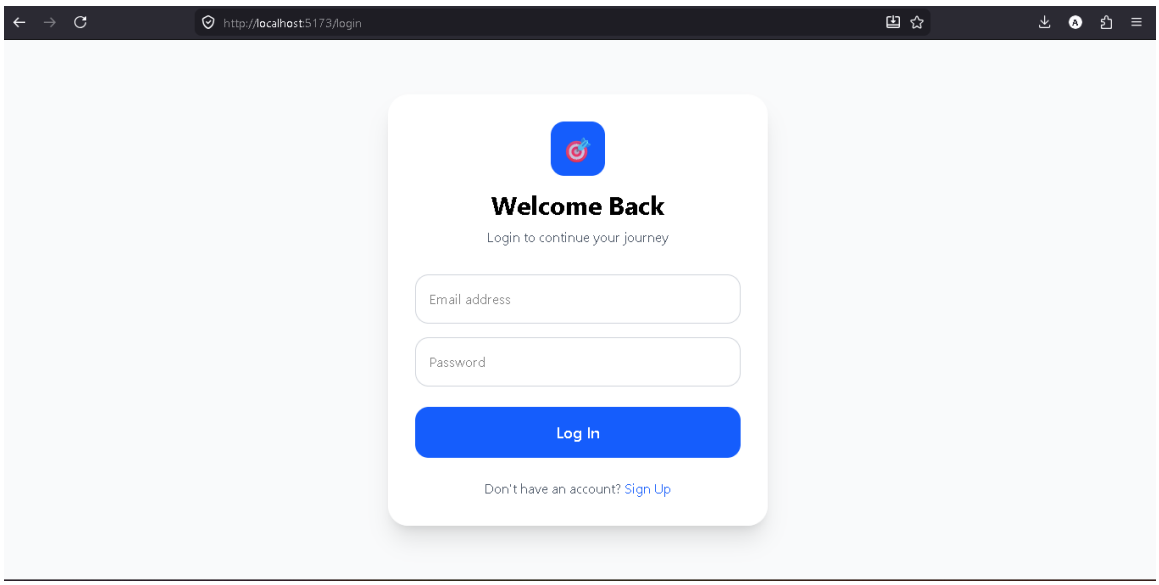

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Figure A.2: CV Upload Page

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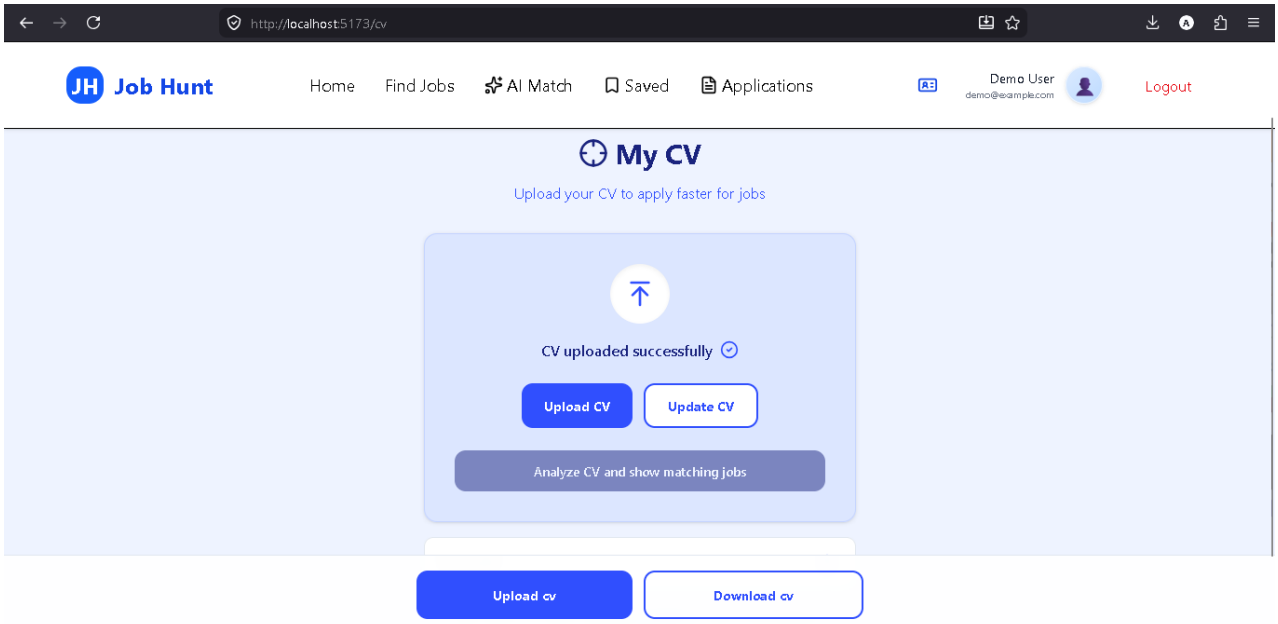

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Figure A.3: AI-Powered Job Recommendations

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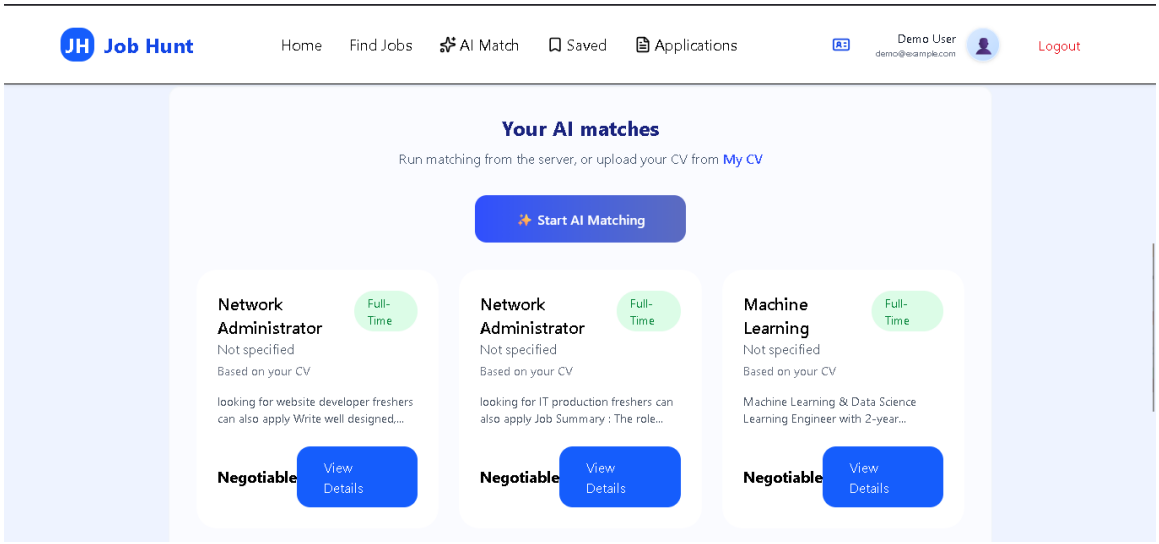

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Figure A.4: Postman Request/Response for /recommend

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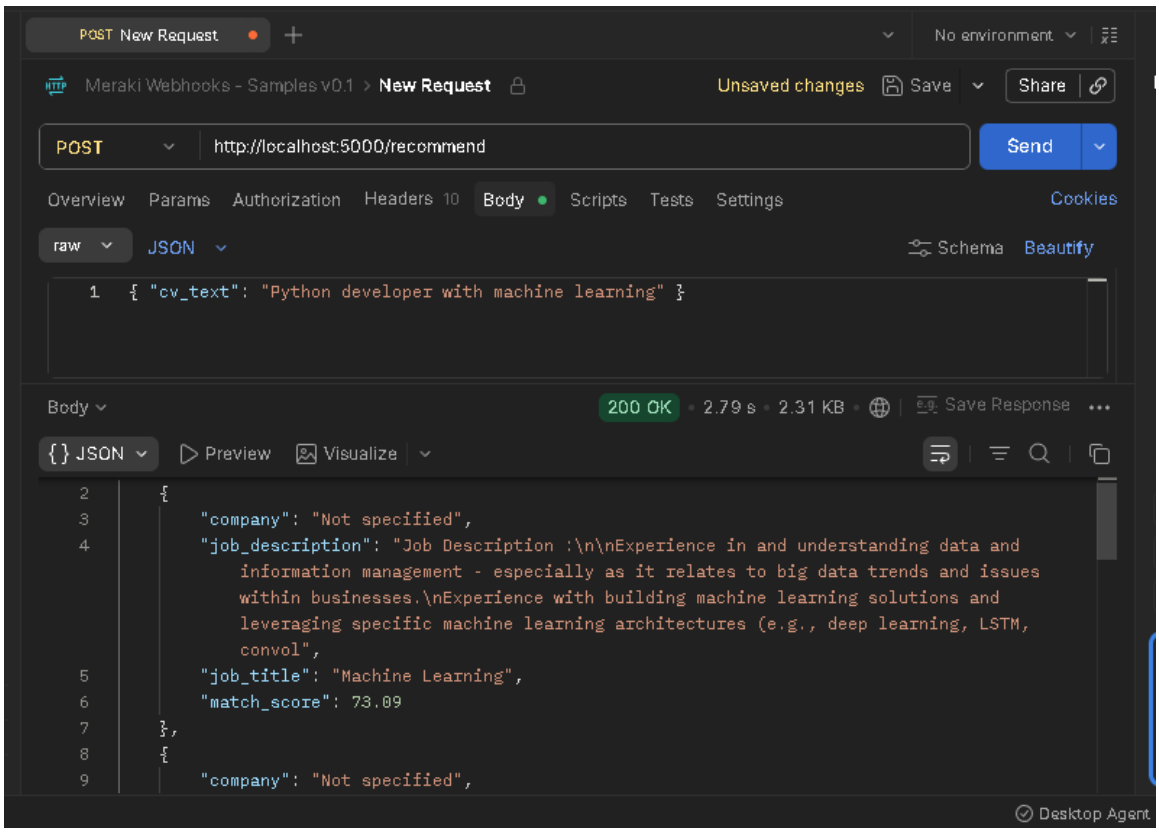

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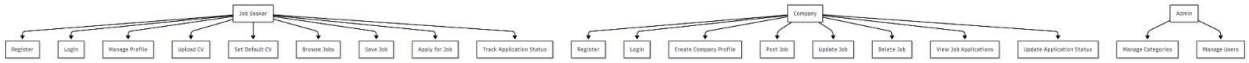
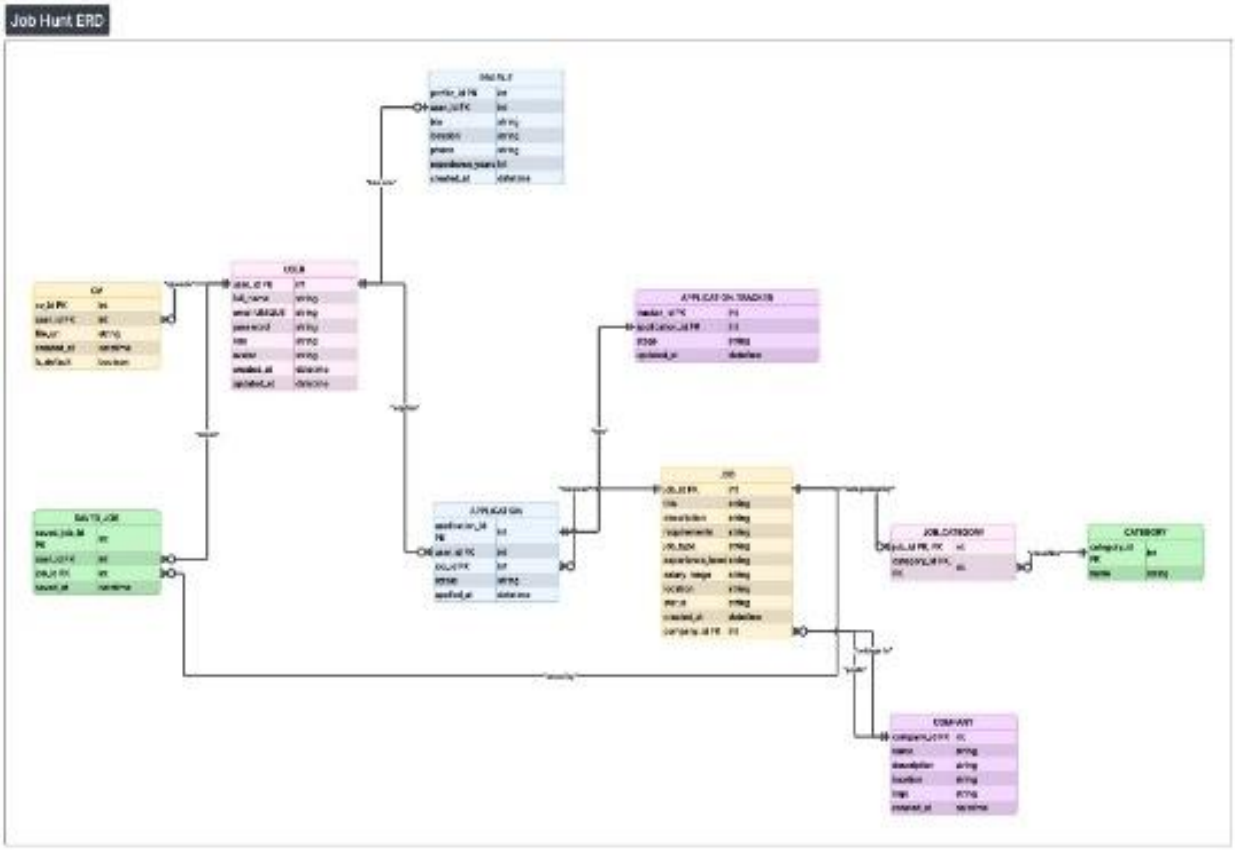
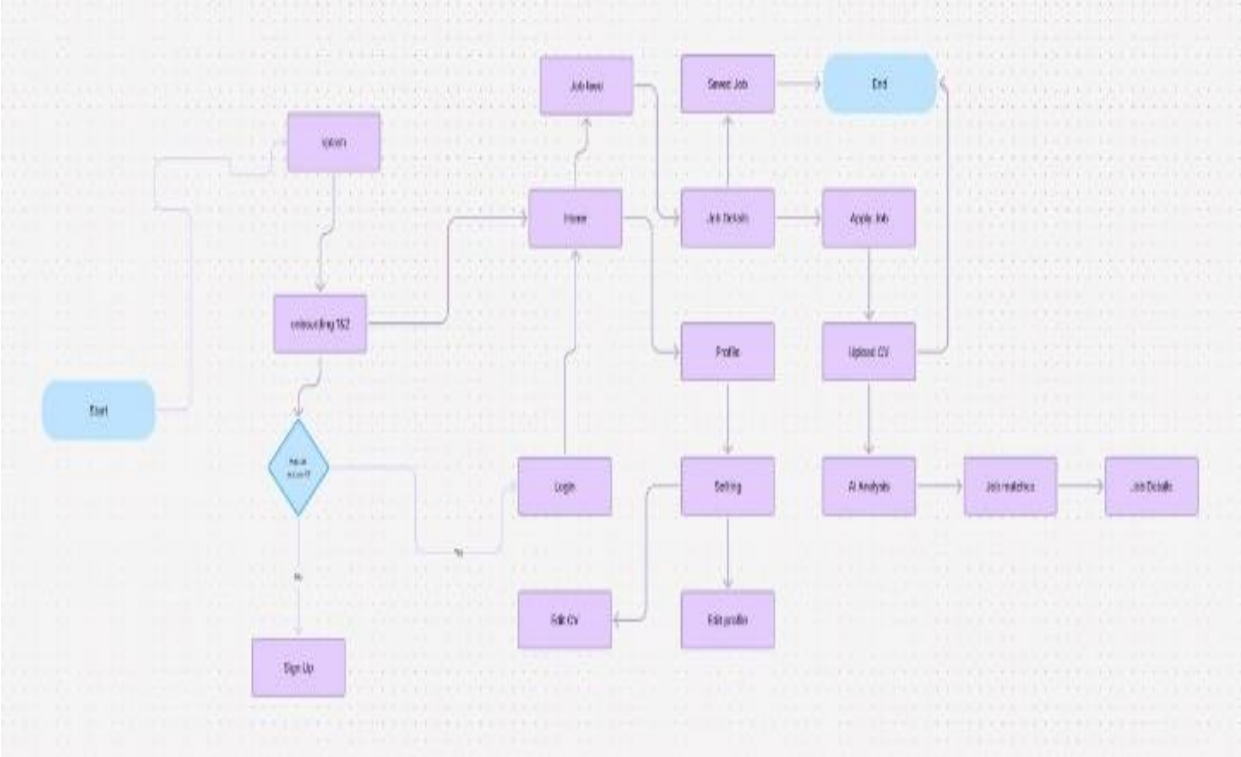
Appendix B: System Diagrams







Appendix C: Sample Code Snippets

C.1 Intelligent Job Matching Algorithm (AI-Based Scoring)

The following code snippet demonstrates the main logic of the intelligent matching system used in *Job Hunt*.
It calculates a matching score between the job seeker profile (skills) and the job requirements, then ranks jobs based on the highest similarity score.

```
def calculate_match_score(user_skills, job_requirements):
    user_skills = set([skill.lower() for skill in user_skills])
    job_requirements = set([req.lower() for req in job_requirements])

    common_skills = user_skills.intersection(job_requirements)

    if len(job_requirements) == 0:
        return 0

    score = (len(common_skills) / len(job_requirements)) * 100
    return round(score, 2)


def recommend_jobs(user_profile, jobs_list):
    recommendations = []

    for job in jobs_list:
        score = calculate_match_score(user_profile["skills"],
job["requirements"])
        recommendations.append({
            "job_title": job["title"],
            "company": job["company"],
            "match_score": score
        })

    recommendations = sorted(recommendations, key=lambda x:
x["match_score"], reverse=True)
    return recommendations[:5]
```

model.py :

```
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics.pairwise import cosine_similarity

import pandas as pd

import re


SKILLS = ["python", "java", "sql", "machine learning", "html", "css",

        "javascript", "react", "django", "flask", "aws", "docker"]


def clean_text(text):

    text = text.lower()

    text = re.sub(r'[^a-zA-Z\s]', '', text)

    return text


def extract_skills(text):

    text = clean_text(text)

    return [skill for skill in SKILLS if skill in text]
```

```
def compute_similarity(cv_text, job_texts):
    vectorizer = TfidfVectorizer()
    vectors = vectorizer.fit_transform([cv_text] + job_texts)
    scores = cosine_similarity(vectors[0], vectors[1:])[0]
    return scores * 100


def recommend_jobs(cv_text, jobs_df):
    job_texts = jobs_df["Job Description"].tolist()
    similarities = compute_similarity(cv_text, job_texts)
    user_skills = extract_skills(cv_text)
    results = []
    for i, row in jobs_df.iterrows():
        job_skills = extract_skills(row["Job Description"])
        skill_match = len(set(user_skills) & set(job_skills))
        skill_score = (skill_match / len(job_skills)) * 100 if job_skills else
0
        final_score = (0.6 * skill_score) + (0.4 * similarities[i])
```

```
        results.append({

            "job_title": row["Job Title"],
            "match_score": round(final_score, 2)
        })
    results.sort(key=lambda x: x["match_score"], reverse=True)
    return results[:5]
```

Appendix E: Installation Guide

Appendix E: Installation Guide for the AI Service

E.1 Prerequisites

- Python 3.10 or higher
- pip package manager
- Git (optional)

E.2 Clone the Repository

```bash
git clone https://github.com/3mroooo/Ai-Project.git
cd Ai-Project/ai_service

E.3 Install Dependencies

bash

```
pip install -r requirements.txt
```

E.4 Run the Flask API

bash

```
python app.py
```

The API will start on `http://localhost:5000`.

Appendix F: Troubleshooting Common Issues

Appendix F: Troubleshooting Common Issues

Issue	Cause	Solution
`ModuleNotFoundError: No module named 'flask'`	Flask not installed	Run `pip install flask`
`Connection refused` when calling API	Flask not running	Start API with `python app.py`
CSV file not found	Wrong file path	Ensure `job_descriptions.csv` exists in the same directory as `app.py`
Recommendations are always the same	Vectorizer fitted only on old data	Restart the API to re-load the CSV

Appendix G: Environment Variables (Optional)

```
Appendix G: Environment Variables

The Flask API supports the following environment
variables:

| Variable | Default | Description |
|----------|---------|-------------|
| `PORT` | 5000 | Port on which the API listens.
|
| `DEBUG` | False | Set to `True` to enable Flask
debug mode. |
| `CSV_PATH` | `job_descriptions.csv` | Custom pa
th to the job descriptions file. |
```

Description:

This AI-based matching algorithm compares the user's skills with job requirements and generates a percentage-based matching score. The system then sorts job opportunities and recommends the top jobs with the highest compatibility.

C.2 Final Recommendation Decision Logic

This snippet combines both skill-based matching and text similarity to generate a final score.

```
def final_matching_score(skill_score,
text_score):
    final_score = (0.6 * skill_score) + (0.4
* text_score)
    return round(final_score, 2)
```

Description

The final score is computed using weighted scoring, where skill matching contributes 60% and text similarity contributes 40%.
This hybrid strategy ensures both technical skill alignment and semantic relevance.

Appendix D: User Manual for the AI Job Match Feature

D.1 Accessing the Feature

After logging in, click on

****"AI Match"** in the navigation bar or**

****"AI Career Assistant"** on the dashboard.**

D.2 Uploading Your CV

1. Click **"Upload CV".**

2. Choose a file (PDF, DOCX, or TXT).

3. Click **"Analyze CV and show matching jobs".**

4. Wait 2-3 seconds.

The system will redirect you to the results page.

D.3 Reading the Results

Each recommendation displays:

- **Job title (click to view full description)**

- **Company name**

- **Match percentage (0–100%)**

- A short excerpt of the job description

A match percentage above **70% indicates a strong fit; **40–70%** indicates moderate relevance; below **40%** suggests low relevance.**

D.4 Applying to a Job

Click on any job card to open its details page, then click **"Apply".**

You may be prompted to confirm your CV and cover letter.

D.5 Troubleshooting

| Problem | Solution |

|---------|----------|

| No results appear | Ensure your CV contains at least 100 characters. Add explicit skill keywords (e.g., “Python”, “SQL”). |

| PDF upload fails | Convert the PDF to a text file (copy-paste) and upload as TXT. |

| Match scores are low | Add more skills and detailed experience descriptions to your CV. |

Conclusion: The Future of Smart Recruitment Starts Here

1. Summary of Achievements

The Job Hunt platform has successfully demonstrated that an intelligent, transparent, and lightweight job recommendation system can be built using open-source technologies and classical NLP techniques (TF-IDF, cosine similarity, and rule-based skill extraction). The system enables users to:

- Upload CVs in multiple formats (PDF, DOCX, TXT).

- Receive ranked job recommendations based on a hybrid scoring model (60% skill matching + 40% semantic similarity).

- Browse job listings, view company details, and apply directly through an intuitive React-based interface.

The AI service, developed as a standalone Flask API, responds in under one second and can be deployed locally or on the cloud. All components are fully documented, and the code is open-source, allowing future developers to extend, modify, or commercialise the platform.

2. Underlying Philosophy: Transparency, Simplicity, and Adaptability

Unlike many commercial job portals that operate as black boxes, Job Hunt was built around three core principles:

- **Transparency: Every match score is explained by the hybrid formula; users know exactly why a job was recommended.**

- **Simplicity: A user does not need to build a detailed profile. Uploading a CV is enough to receive personalised recommendations.**

- **Adaptability: The system is modular. The skill list, the hybrid weight, and even the vectorization algorithm can be changed without rewriting the entire codebase.**

These principles make Job Hunt suitable not only for fresh graduates but also for academic research in recommendation systems and for small businesses that need an affordable, customisable recruitment tool.

3. How to Use Job Hunt (End-User Perspective)

For a first-time user, the workflow is straightforward:

1. **Register / Log in to the web application.**

2. **Upload your CV (PDF, DOCX, or TXT) on the "My CV" page.**

3. **Click "Analyze CV" – the system will extract your skills and compare them with the job database.**

4. **View your top-5 job matches on the "AI Match" page. Each recommendation includes the job title, company name, match percentage, and a short description.**

5. **Apply directly by clicking "View Details" and then "Apply".**

No machine learning expertise is required. The platform is designed for end-users who only want to find relevant jobs quickly.

4. Societal Impact and Alignment with Egypt's Vision 2030

Graduate unemployment remains a critical challenge in Egypt and the broader MENA region. Traditional job search methods (newspaper ads, generic job boards) often fail to match fresh graduates with suitable entry-level positions. By providing a free, transparent, and AI-powered recommendation engine, Job Hunt contributes to:

- **Reducing the time and effort required for job hunting.**

- **Empowering students and recent graduates with actionable feedback on their CVs.**

- **Helping small and medium enterprises find qualified candidates without expensive recruitment software.**

Moreover, the platform aligns with three United Nations Sustainable Development Goals (SDGs):

- **SDG 8 (Decent Work and Economic Growth): Promotes employment for young people.**

- **SDG 9 (Industry, Innovation, and Infrastructure): Uses AI to modernise the recruitment industry.**

- **SDG 10 (Reduced Inequalities):** Offers equal access to job opportunities regardless of the user's background.

5. Closing Statement

Job Hunt is more than a graduation project; it is a proof of concept that transparent, lightweight, and user-centric AI can be applied to solve a real-world problem. It demonstrates that classical algorithms (TF-IDF, cosine similarity), when combined with a well-designed hybrid scoring model, can still outperform opaque, heavyweight solutions in specific domains.

The platform is now ready for demonstration, testing, and future extension. Whether used by a student looking for their first internship or by a small company seeking a customisable recruitment tool, Job Hunt delivers on its promise: **smart, fast, and fair job recommendations for everyone.**

"In a world of black-box algorithms, Job Hunt shows that the best recommendation is an explained one."

Note :

The matching system in Job Hunt is designed to be scalable and adaptable.
Weights can be adjusted based on future improvements, such as adding deep learning models or user behavior tracking.